SHORTEST PATH ANALYSIS FOR DIALYSIS CENTER PLACEMENT IN NATONIN, MOUNTAIN PROVINCE

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1 ABSTRACT

The aim of this study is to identify the optimal location for a new dialysis center in Natonin, Mountain Province, Philippines. The researchers used graph theory concepts such as eccentricity and the Bellman-Ford algorithm to calculate the distances between different barangays and existing dialysis centers. The study also determined the shortest routes between barangays and existing dialysis centers, as well as the shortest routes from the proposed dialysis center location to each barangay. The findings indicate that Barangay Poblacion is the optimal location for the new dialysis center due to its central location and proximity to all other barangays. Establishing a dialysis clinic in this area will significantly decrease patient travel lengths, increase accessibility, and improve local healthcare services.

Keywords— Bellman-Ford algorithm, eccentricity, shortest route, dialysis center

2 INTRODUCTION

2.1 Background of the Study

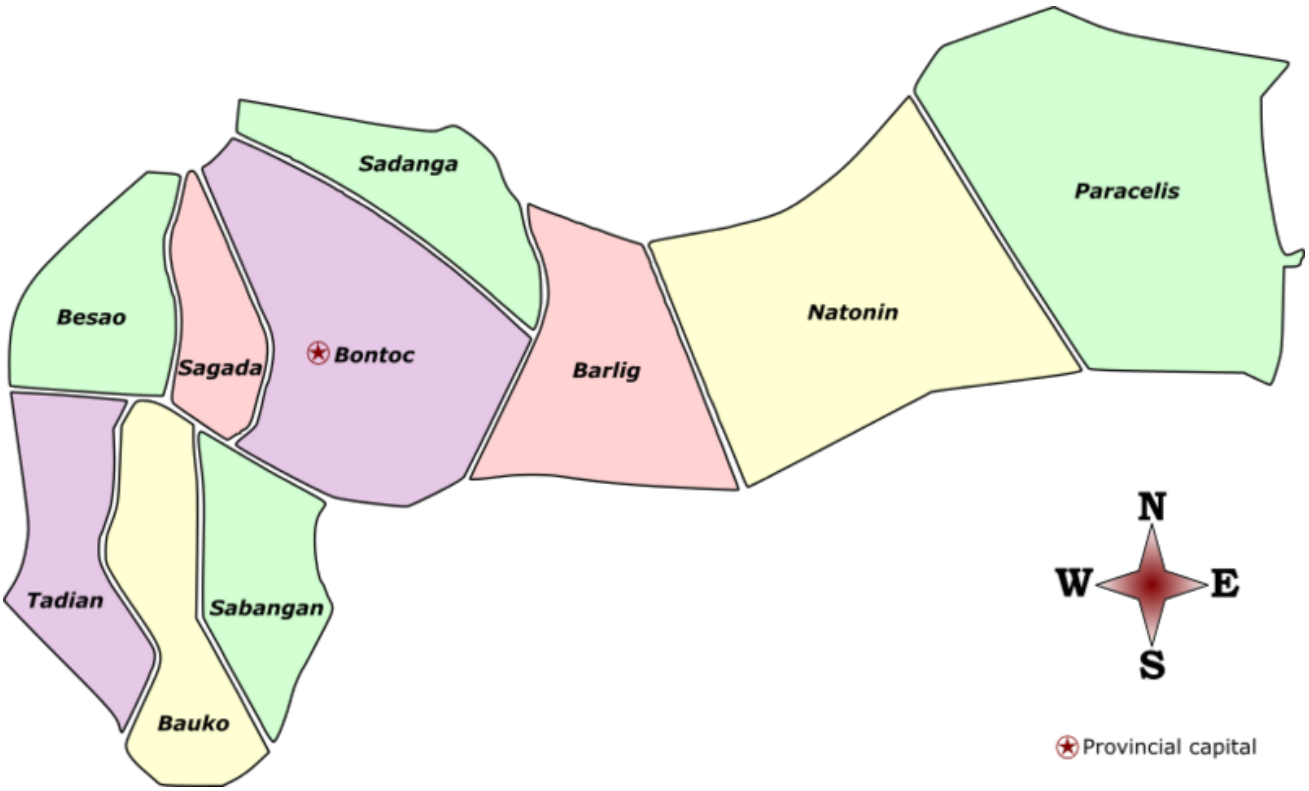
Kidneys act as a natural filter of our body that cleans the bloodstream by removing waste in a form of urine. The National Cancer Institute also stated that kidneys help keep chemicals such as sodium, potassium, and calcium balanced in the body to maintain equilibrium. Thus, having one or both kidneys fail is a life-threatening condition since the bloodstream cannot be filtered anymore and harmful chemicals such as ammonia and urea may accumulate in the bloodstream, causing complications. If a kidney loses $85-90\%$ of its function, it implies that the person has kidney failure and must undergo dialysis or kidney transplant to survive, as stated by the National Kidney Foundation (2023) in New York.

Hemodialysis is a life-saving treatment for people with kidney failure that allows the body to eliminate excess fluid and waste products from the blood when the kidneys are unable to do so (National Kidney Foundation, 2024). A hemodialysis machine acts as an artificial

kidney, eliminating waste and excess fluid from the body to prevent accumulation. It also maintains healthy levels of minerals in the blood, including potassium, sodium, calcium, and bicarbonate, and regulates blood pressure. Additionally, it was stated that hemodialysis can be performed in a hospital, a dialysis center, or at home. Treatments in a dialysis center are usually performed three times a week, each taking 3 to 4 hours to complete.

In an interview with Dr. Romina Danguilan, Deputy Executive Director for Medical Service of National Kidney and Transplant Institute, the number of new dialysis patients in the Philippines increased by more than 40% between 2022 and 2023 (Cabato, 2024). The Cordillera Administrative Region (CAR) has seen a considerable increase in the frequency of chronic kidney disease (CKD), resulting in a rise in the number of dialysis patients (World Health Organization, 2013). This concerning trend is worsened by the limited access of the region to specialized healthcare services, especially in rural areas. The growing number of dialysis patients calls for the establishment of more dialysis centers. In March 2024, the Philippine Health insurance Corporation reported thirteen accredited independent dialysis clinics in CAR; one in Abra, nine in Benguet, one in Ifugao, and two in Kalinga. Bontoc General Hospital, located in the capital of Mountain Province, also offers dialysis treatments.

According to the Department of Health (2015), Mountain Province, like many other rural areas in the Philippines, has seen an increase in noncommunicable diseases, especially kidney disease. The scarcity of specialist care, combined with the difficulties of accessing healthcare in distant areas, can lead to higher morbidity and death among patients with kidney diseases. This growth has emphasized the crucial need for improved dialysis services in the region, particularly for patients living in distant locations. Mountain Province has ten municipalities where Bontoc is the capital of the Province (Figure 1).



Source: Wikipedia

Figure 1: Map of Mountain Province

Researchers of this study from Mountain Province or have previous experience in Mountain Province stated that Natonin and Paracelis are distant from the capital. Traveling from Natonin or Paracelis to Bontoc is challenging since, aside from the distance, some roads and pathways are in rough condition. Natonin has seen an increase in demand for dialysis treatments, particularly among those living in remote locations and facing limited resources. Although specific data on the increase in dialysis patients in Natonin may be limited, the overall national trend of increased CKD prevalence indicates a growing demand for such treatments (Tang, et. al., 2020). Barangay health stations are established to address the medical services needed for the city; however, it does not support dialysis due to lack of equipment. Patients would either go to Kalinga for dialysis treatment as it is relatively nearer than Bontoc.

2.2 Importance of the Study

Natonin is a municipality of Mountain Province with eleven barangays namely Alunogan, Balangao, Banao, Banawal, Butac, Maducayan, Poblacion, Pudo, Saliok, Sta. Isabel, and Tonglayan. It confronts unique obstacles in providing proper medical care to its residents due to its geographic isolation and limited infrastructure. The rocky terrain, steep slopes, and lack of paved roads make it difficult to reach healthcare facilities, especially for individuals living in remote barangays.

Source: Google Maps

Figure 2: Satellite Map of Natonin, Mt. Province

The emergence of a new dialysis clinic in Natonin would be a critical step towards addressing gaps in healthcare. By carefully placing the facility, we can increase accessibility, reduce patient travel distances, and ensure that the benefits of dialysis services flow effectively throughout the community especially those in outlying barangays. It can help in reducing overcrowding in current dialysis facilities and improving overall service delivery. In addition,

the new dialysis center will also cater near cities such as Paracelis and barangays of Bontoc near Natonin. Presenting the shortest route from the dialysis center to each barangay would also help local government units to make informed decisions in establishing the facility to the proposed location. The optimal routes that will be presented may be of use especially of private vehicles or ambulances to effectively transport the patient especially in times of emergency.

2.3 Objectives of the Study

The aim of this study is to determine the best location for a new dialysis center in Natonin, Mountain Province, using geographical parameters such as distance and road network connectivity. Additionaly, the study will aim to

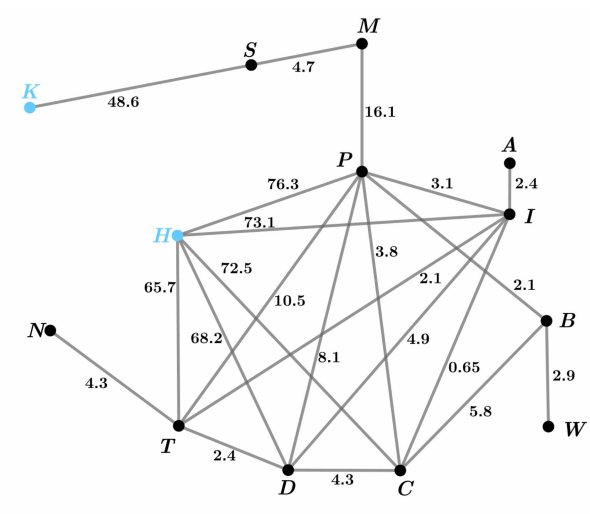
1. Identify the shortest route between the barangays of Natonin and existing medical facility offering dialysis;
2. Determine the most suitable location for a new dialysis center; and
3. Present the shortest route from the proposed location of dialysis center to each barangay in Natonin.

3 METHODOLOGY

To identify the area in Natonin, Mountain Province that can accommodate the dialysis center, the researchers studied the districts in the municipality and the existing hospitals that offer dialysis treatment near it. They applied concepts in graph theory such as eccentricity to identify the district that is most central to the city. They next applied the Bellman-Ford algorithm to determine the nearest routes from this point to all the other districts. Each of the districts and dialysis centers were designated as a node and the distance between them as the weight. These distances were estimated using Google Maps.

The hospitals chosen were Bontoc General Hospital as it is within the province and Kalinga Provincial Hospital situated in Tabuk, Kalinga. The two existing dialysis centers

were disregarded as both centers are near the provincial hospital. The map of Natonin was modeled as a graph (refer to Figure 3) with nodes A, B, C, D, H, I, K, M, N, P, S, T, and W representing Alunogan, Balangao, Butac, Pudo, Bontoc General Hospital, Sta. Isabela, Kalinga Provincial Hospital, Maducayan, Banao, Poblacion, Saliok, Tonglayan, and Banawal respectively. The graph was verified using Google Maps to confirm existence of passable roads or pathways. One of the researchers from Mountain Province also verified the edges of the graph. The point of reference for each barangay are the barangay health stations as these are the nearest medical facility to the barangays and there are more neighboring households. Due to the absence of barangay health stations in Brgy. Poblacion, Brgy. Banao, and Brgy. Saliok, barangay halls are chosen for the same reason. Elementary schools are chosen points of reference for Brgy. Pudo and Brgy. Maducayan.



Node	Representation
A	Alunogan
B	Balangao
C	Butac
D	Pudo
H	Bontoc General Hospital
I	Sta. Isabela
K	Kalinga Provincial Hospital
M	Maducayan
N	Banao
P	Poblacion
S	Saliok
T	Tonglayan
W	Banawal

Figure 3: Barangays of Natonin, Mt. Province

The graph above (drawn not to scale) shows the location of different districts in Natonin and the distances between them and together with a summary of the representation of each

node. The nodes highlighted in blue are the hospitals offering dialysis treatment near the area.

3.1 Eccentricity of a Graph

The eccentricity of a vertex u in a connected graph G is the distance between u to the furthest vertex v to u in G (Chartrand et al., 2016). It is calculated by finding the shortest path between the vertex and every other vertex and then selecting the longest distance. For example to find the eccentricity of node K, its distance (regardless of weights) of K to every other node of the graph will be considered. Weights are disregarded so that the structure of the city will be more focused since the researchers are more concerned to the structural relationship between vertices.

Node	**Place**	**Path**	**Distance (number of edges)**
A	Alunogan	$K \to S \to M \to P \to I \to A$	5
B	Balangao	$K \to S \to M \to P \to B$	4
C	Butac	$K \to S \to M \to P \to C$	4
D	Pudo	$K \to S \to M \to P \to D$	4
H	Bontoc General Hospital	$K \to S \to M \to P \to H$	4
I	Sta. Isabela	$K \to S \to M \to P \to I$	4
M	Maducayan	$K \to S \to M$	2
N	Banao	$K \to S \to M \to P \to T \to N$	5
P	Poblacion	$K \to S \to M \to P$	3
S	Saliok	$K \to S$	1
T	Tonglayan	$K \to S \to M \to P \to T$	4
W	Banawal	$K \to S \to M \to P \to B \to W$	5
		Eccentricity	5

Table 1: Eccentricity of Kalinga Provincial Hospital (K)

The table above shows the distance of node K (Kalinga Provincial Hospital) to every other nodes in the graph. It can be seen that the furthest distance node to K are A (Alunogan), N (Banao), and Banawal (W). Thus, it can be said that the eccentricity of node K is 6. To solve for the eccentricity of every other nodes, same procedure will be used.

3.2 Bellman-Ford Algorithm

The Bellman-Ford algorithm is a graph search algorithm that computes shortest paths from a single source vertex to all other vertices in a weighted graph. It is a dynamic programming algorithm that works by iteratively relaxing the edges of the graph until it finds the shortest path from the source vertex to all other vertices. It can handle negative edge weights, but it cannot handle graphs with negative cycles.

The central node of the graph will be the single source vertex point to all other vertex. Node H (Bontoc General Hospital) and K (Kalinga Provincial Hospital) were also considered as the starting point vertex going to every barangay of Natonin for comparison purposes. The shortest paths are computed from central node or H or K to every other vertex of the graph by checking if the distance from the source to the destination vertex can be shortened by gradually exploring the roads, and updating the distances to destinations if there exists a shorter path.

4 RESULTS AND DISCUSSION

4.1 Shortest Route from Hospitals

Barangay	**Path**	**Total Distance (km)**
Alunogan	$H \to T \to D \to I \to A$	75.40
Balangao	$H \to T \to D \to C \to B$	78.20
Banao	$H \to T \to N$	70.00
Banawal	$H \to T \to D \to C \to B \to W$	81.10
Butac	$H \to T \to D \to C$	72.40
Maducayan	$H \to T \to D \to I \to P \to M$	92.20
Poblacion	$H \to T \to D \to I \to P$	76.10
Pudo	$H \to T \to D$	68.10
Saliok	$H \to T \to D \to I \to P \to M \to S$	96.90
Sta. Isabel	$H \to T \to D \to I$	73.00
Tonglayan	$H \to T$	65.70

Table 2: Shortest Route from Bontoc General Hospital (H)

Table 2 presents the outcomes obtained from the algorithm used to determine the distance between the barangays and Bontoc General Hospital. While certain barangays like Brgy. Tonglayan and Brgy. Pudo are relatively near to the hospital, Brgy. Saliok and Brgy. Maducayan is located at a great distance, making it exhausting for patients to go to the hospital for dialysis treatments. A patient who lives in Natonin who needs dialysis treatment in Bontoc General Hospital will travel at a distance between 65.70 – 96.90 km.

Barangay	**Path**	**Total Distance (km)**
Alunogan	$K \to S \to M \to P \to I \to A$	74.90
Balangao	$K \to S \to M \to P \to B$	71.50
Banao	$K \to S \to M \to P \to I \to D \to T \to N$	84.10
Banawal	$K \to S \to M \to P \to B \to W$	74.40
Butac	$K \to S \to M \to P \to I \to C$	73.15
Maducayan	$K \to S \to M$	53.30
Poblacion	$K \to S \to M \to P$	69.40
Pudo	$K \to S \to M \to P \to I \to D$	77.40
Saliok	$K \to S$	48.60
Sta. Isabel	$K \to S \to M \to P \to I$	72.50
Tonglayan	$K \to S \to M \to P \to I \to D \to T$	79.80

Table 3: Shortest Route from Kalinga Provincial Hospital (K)

The results in Table 3 presented the paths and distance from Kalinga Provincial Hospital to every barangay in Natonin. For this result, Brgy. Saliok and Brgy. Maducayan are the nearest barangay to the hospital while Brgy. Banao and Brgy. Tonglayan are the farthest. A patient who lives in Natonin who needs dialysis treatment in Bontoc General Hospital will travel at a distance between 48.60 – 84.10 km.

Barangay	Distance	
	Kalinga Provincial Hospital	*Bontoc General Hospital*
Alunogan	74.90	75.40
Balangao	71.50	78.20
Banao	84.10	70.00
Banawal	74.40	81.10
Butac	73.15	72.40
Maducayan	53.50	92.20
Poblacion	69.40	76.10
Pudo	77.40	68.10
Saliok	48.60	96.90
Sta. Isabel	72.50	73.00
Tonglayan	79.80	65.70

Table 4: Comparison of Distance

The numbers colored in blue are the minimum distance among the hospitals with respect to the barangay. Most barangays are relatively near to Kalinga Provincial Hospital than in Bontoc General Hospital. The hospital is in the center of Mountain Province, while Natonin is close the province's border. Thus, it is more convenient to seek dialysis services in another province, such as Kalinga in terms of distance. This explains why patients would go to Kalinga for treatment instead of going to Bontoc.

4.2 Shortest Route including Proposed Location of the Dialysis Center

4.2.1 *Strategic Placement of the Dialysis Center*

Figure 4 shows the eccentricity of each node of the graph. The nodes with minimum eccentricity of 3 are nodes M and P which represents Brgy. Macudayan and Brgy. Poblacion respectively. These two are then considered as the central nodes of the graph or the most centrally located vertex in the graph. This implies that nodes P and M are closest to all other vertices in the graph.

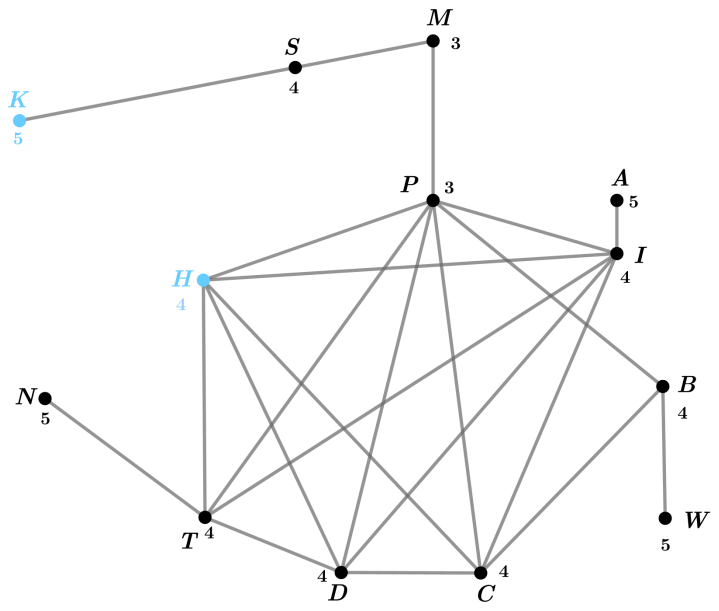


Figure 4: Eccentricity of each node

Based from the paths presentend in Table 3, Brgy. Maducayan always passes Brgy. Poblacion when traveling from Kalinga Provincial Hospital to every barangay. Same behavior can be observed in Table 2 where a Brgy. Saliok and Brgy. Maducayan always passes Brgy. Poblacion to go to Bontoc General Hospital. With this, the researchers inferred that Brgy. Poblacion be the center of the city. This barangay would also be the proposed location to build a dialysis center as it is relatively near to all barangays in Natonin.

4.2.2 *Shortest Route From the Dialysis Center*

From the results of the eccentricity in Figure 4, Brgy. Poblacion was used as the proposed location of the dialysis center and source node for the algorithm to determine the shortest route between the center and every barangay in Natonin. Table 5 summarizes the result from the algorithm showing the paths and distances to each barangay. The range of distance that a patient will travel was minimized between 2.10 – 20.80 km.

Barangay	**Path**	**Distance (km)**
Alunogan	$P \to I \to A$	5.50
Balangao	$P \to B$	2.10
Banao	$P \to I \to D \to T \to N$	14.70
Banawal	$P \to B \to W \to S$	5.00
Butac	$P \to I \to C$	3.75
Maducayan	$P \to M$	16.10
Poblacion	P	0.00
Pudo	$P \to I \to D$	8.00
Saliok	$P \to M \to S$	20.80
Sta. Isabel	$P \to I$	3.10
Tonglayan	$P \to I \to D \to T$	10.40

Table 5: Shortest Route from Brgy. Poblacion (P)

Table 6 is the updated summary of Table 4 where the proposed location of the dialysis center was added to the comparison of distances from the barangays to the hospitals and center offering dialysis services. The numbers colored in blue are the minimum distance of the barangays to the existing hospitals while the numbers in green are the minimum distance between considering Brgy. Poblacion.

Barangay	**Distance**		
	Kalinga Provincial Hospital	*Bontoc General Hospital*	*Barangay Poblacion*
Alunogan	74.90	75.40	5.50
Balangao	71.50	78.20	2.10
Banao	84.10	70.00	14.70
Banawal	74.40	81.10	5.00
Butac	73.15	72.40	3.75
Maducayan	53.50	92.20	16.10
Poblacion	69.40	76.10	0.00
Pudo	77.40	68.10	8.00
Saliok	48.60	96.90	20.80
Sta. Isabel	72.50	73.00	3.10
Tonglayan	79.80	65.70	10.40

Table 6: Comparison of Distance including Brgy. Poblacion

The additional dialysis center from Brgy. Poblacion greatly reduces the distance traveled by the patients in Natonin. Brgy. Poblacion is also located near the highway thus it can be easily accessed by the residents of Natonin. Adding a dialysis center in this barangay will

also reduce the travel time of patients especially those with limited mobility and those who will come from barangays with transportation difficulties (e.g., poor road conditions). This is also crucial in case of medical emergencies where dialysis is needed as soon as possible (e.g., uremia, electrolyte imbalance, edema).

Overall, healthcare services should be accessible to all individuals across all cities and provinces, regardless of their location, economic circumstances, or the challenges posed by road conditions or limited resources. Dialysis is a major part of someone's life and ensuring access to this life-saving treatment is a responsibility that the government must fulfill.

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